

FINANCIAL MATH & ANALYSIS CONCEPTS (190)

—OPEN EVENT—

NATIONAL – 2019

DO NOT WRITE ON TEST BOOKLET

TOTAL POINTS _____ (100)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than 60 minutes testing time

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GENERAL INSTRUCTIONS

The number of decimal places used in calculations and in the reporting of your answers is very important. Below are guidelines to use as you take this test. If a problem varies from these guidelines, it will be noted in the problem.

PAYROLL QUESTIONS

- Do not round the overtime rate.
- Overtime is paid for any hours over 40 worked in a week, *unless otherwise stated in the problem.*
- Normal work week is 40 hours, *unless otherwise stated in the problem.*

PERCENTAGE PROBLEMS

- Round to the nearest tenth when working with percentages.

Example: Use 2.3% NOT 2%

Example: An answer of .9638 should be written as 96.4%

FRACTIONS

- All problems with fractions are to be reduced to their lowest terms.

INTEREST

- All interest rates are stated in one-year simple interest, *unless otherwise noted.*
- Interest is to be calculated on 365 days.

DOLLAR AMOUNTS

- All dollar amounts must be rounded to the second place (nearest cent).

Example: \$35.9765 to \$35.98

1. The bank statement sent to Ann Ochocink did *not* show checks for \$158.23, \$12.89, and \$71.27, and an outstanding deposit of \$75 that were listed in the check register. The balance printed on the statement was \$289.07. What is the reconciled bank statement balance?
 - a. (\$28.32)
 - b. \$121.68
 - c. \$456.46
 - d. \$606.46
2. Claire bought a used car for \$13,576. Her expenses for the first year were gas and oil, \$900; repairs, \$322; insurance, \$573; license plates, \$54.25; loss of interest on her car investment, \$611; and depreciation, 11%. Find the total operating cost for the year, rounded to the nearest dollar.
 - a. \$2,460
 - b. \$3,954
 - c. \$9,622
 - d. \$11,116
3. Which of these is *not* a benefit included under the FICA tax?
 - a. disability benefits
 - b. Medicare
 - c. fringe benefits
 - d. retirement benefits
4. A cupcake shop has current assets of \$60,100 and current liabilities of \$32,100. Which of these *best* represents the shops current ratio?
 - a. 28,000
 - b. 2:1
 - c. 1.9:1
 - d. 0.53:1
5. One quart is equivalent to 32 oz., which is equivalent to 0.95 L. How many liters are in a can that holds 48 oz. of paint thinner?
 - a. 1.9 L
 - b. 1.579 L
 - c. 1.425 L
 - d. 0.633 L
6. Syringa Wireless Company sold 22,512 handheld units last year. This year the new phones have been reduced in size by 20%. The company expects sales to increase 35% from last year's sales. How many unit sales is Syringa forecasting for this year?
 - a. 7,879
 - b. 8,010
 - c. 27,014
 - d. 30,391

7. After the local surf and turf restaurant went bankrupt, the trustee sold the firm's assets for \$156,400. The legal costs of the bankruptcy and other claims that had to be paid before creditors totaled \$68,800. Creditor's claims totaled \$212,600. How much will a creditor get who has a claim of \$43,100?
- \$17,759
 - \$31,707
 - \$43,100
 - \$45,654
8. Typically a piece-rate employee is paid only for
- the total number of rejected pieces produced
 - the total number of usable pieces produced
 - the total number of usable pieces produced plus the number of hours worked
 - none of the above
9. Which of the following describes how to calculate net pay?
- Subtract the total deductions from the net pay.
 - Add the total deductions to the gross pay.
 - Subtract the total deductions from the gross pay.
 - Subtract the total deductions from the take-home pay.
10. You are thinking of buying a home for \$104,300 and making an 18% down payment. You estimate closing costs will be 2.5% of the home's purchase price, which must be paid in cash at the time of closing. How much cash will you need to buy the home?
- \$16,166.50
 - \$18,744
 - \$19,243.35
 - \$21,381.50
11. A(n) _____ is an amount of income per person that is free from tax.
- exemption
 - state tax
 - inheritance
 - part-time job for a student
12. Which of the following describes how to calculate simple interest?
- $\text{Interest} = \text{Principal} \div \text{Rate} \times \text{Time}$
 - $\text{Interest} = \text{Principal} \div \text{Rate} \div \text{Time}$
 - $\text{Interest} = \text{Principal} \times \text{Rate} \div \text{Time}$
 - $\text{Interest} = \text{Principal} \times \text{Rate} \times \text{Time}$

13. Which of the following reasons *might* be used to discourage a friend from getting a cash advance on a credit card?
- Interest charges begin accumulating immediately.
 - Payments are applied to any purchase balance before the cash advance balance creating higher finance charges.
 - The interest rate for cash advances is higher than for purchases.
 - All of the above.
14. How many days are considered to be in a year according to the ordinary interest method?
- 365
 - 366
 - 360
 - 355
15. All of the following are key cost items in the purchase of a new car *except*
- Rebates.
 - Sales tax.
 - Down payment.
 - Lease terms.
16. Penn Williams purchased 300 shares of a stock selling for \$19.74 a share. His broker charged him 3.5% in commission. What was the total cost of the stock to Penn?
- \$6,137.75
 - \$6,129.27
 - \$6,135.92
 - \$213.19
17. Karissa was injured in an accident and can no longer work. Her disability policy will pay 55% of her average annual wages for the last 3 years. Her wages for the last 3 years were \$55,650, \$56,900, and \$58,220. What is Karissa's monthly disability benefit?
- \$31,307.83
 - 170,770.00
 - \$2,608.99
 - \$2,134.63
18. Stratton and Toevs invested \$140,000 and \$360,000, respectively in a partnership. They agreed to share net income in proportion to their investments. If the partnership's net income last year was \$43,500, how much was Toevs' net income?
- \$31,320.00
 - \$12,180.00
 - \$155357.10
 - \$60,416.67

19. To ship freight to destination zone 7, a shipping company charges the following per pound rates based on weight: 1-99 pounds, \$1.14; 100-249 pounds, \$1.05; 250-499 pounds, \$0.98; and 500 pounds and over, \$0.95. What is the freight charge to ship a 230-pound package to zone 7?
- \$270.18
 - \$250.41
 - \$241.50
 - \$248.85
20. A human resource department spent the following amounts of time in hiring a technician: 0.7 hours reviewing the application, 1.8 hours checking references, 0.8 hours interviewing, and 2.2 hours approving the budget. These costs were charged at \$220 an hour. Overhead costs of adding a new technician are \$450. What was the total cost of hiring the technician?
- \$670
 - \$1,210
 - \$1,560
 - \$1,660
21. The Community Grocery Warehouse inventory of cases of 12-ounce cans of baked beans, 12 cans to a case, states the following:

On hand September 1	27 Cases
Sales Week 1	10 Cases
Received Week 2	24 Cases
Sales Week 2	20 Cases
Sales Week 3	13 Cases
Received Week 3	66 Cases
Sales Week 4	30 Cases

How many cases of baked beans does the grocery have on hand on October 1?

- 44 Cases
- 117 Cases
- 56 Cases
- 725 Cases

22. Thors Antiques estimates the annual cost of carrying inventory to be 25 percent. The company usually has an inventory of \$380,000. With the advent of on-time shipping, Thors was able to reduce the amount of inventory it carried to three-fourths of the usual amount. What is Thors' cost of carrying inventory as a result?
- a. \$28,500
 - b. \$52,750
 - c. \$71,250
 - d. \$74,500
23. Rodriguez's Grocery installed new barcode scanning equipment to increase control of inventory. Joaquin Ramirez supervised the installation and handled all the on-site programming for \$2,780. He also provided training to Rodriguez's employees in seven 2-hour sessions at \$225 per session. What is the total cost of Ramirez's professional services?
- a. \$2,780
 - b. \$4,355
 - c. \$1,575
 - d. \$3,005
24. Copies On Demand recently purchased 5 new copiers at a total cost of \$45,000. The salvage value of the copiers is expected to be \$700 each after 5 years of use. Using the straight-line method, what is the total annual depreciation for the first year?
- a. \$8,300
 - b. \$7,585
 - c. \$9,333
 - d. \$8,021
25. Kahn's Snack Shop had a beginning inventory of \$15,700. During the quarter, three shipments received were valued at \$2,375, \$1,402, and \$4,568. The ending inventory for the quarter was valued at \$14,892. Find the cost of goods sold.
- a. \$8,235
 - b. \$9,531
 - c. \$9,153
 - d. \$8,968
26. Gross profit on sales in March was \$2,786. Gross profit on sales in April was \$2,900. What is the percent change?
- a. 1.4 percent
 - b. 2.9 percent
 - c. 3.9 percent
 - d. 4.1 percent

Use the following information to answer questions 27-30.

Drake Insurance is planning a major expansion program. To finance the program, Drake plans to sell an issue of 25,000 shares of stock at \$9.20 per share. The underwriting commission will be 6.5 percent of the value of the stocks. Accounting fees, legal fees, printing costs, and other expenses are estimated to be \$1,985.

27. Find the value of issue.
- a. \$25,000
 - b. \$230,000
 - c. \$14,950
 - d. \$130,000
28. Find the total selling expenses.
- a. \$16,935
 - b. \$1,985
 - c. \$14,950
 - d. \$12,965
29. Find the net proceeds.
- a. \$213,065
 - b. \$230,000
 - c. \$16,935
 - d. \$14,950
30. Find the selling expense per share.
- a. \$6.77
 - b. \$0.68
 - c. \$0.60
 - d. \$6.00
31. Find the maturity value of a loan: \$80,000 borrowed at 3 percent over prime with ordinary interest for 180 days. The prime rate is 5.75 percent.
- a. \$76,000
 - b. \$83,500
 - c. \$76,500
 - d. \$80,000

32. The original price of a box of laundry detergent was \$7.75. The inflation rate was 8.5 percent. Find the current price.
- a. \$8.75
 - b. \$8.14
 - c. \$8.48
 - d. \$8.41
33. On September 1 Redwing Nursery started selling bird seed. It started with 150 bags of sunflower seed in stock. At the end of each week in September, Redwing had sold 37, 62, 75, and 145 bags of seed. During the month, shipments of 122, 85, and 55 bags of seed were received. How many bags of sunflower seed does Redwing have in inventory at the end of September?
- a. 412
 - b. 319
 - c. 159
 - d. 93
34. Rodriguez's Grocery needs 1,200 more square feet of unrefrigerated storage space in its distribution center. The store is willing to pay \$9,000 a year for this space. What is the annual rental rate per square foot Rodriguez's is willing to pay?
- a. \$9.00
 - b. \$7.50
 - c. \$6.00
 - d. \$4.50
35. M's Janitorial Service cleans 4,225 square feet per floor and 32 stories of office space. M's rate for janitorial services is \$5.35 per square foot annually. To the nearest cent, what is M's annual charge for cleaning this office building?
- a. \$135,200.04
 - b. \$271,245.00
 - c. \$361,660.08
 - d. \$723,320.00
36. Stan Timon is an attorney hired by Chin Lee Florist to represent the company in a law suit with one of its suppliers. Timon charges \$125 per hour plus 15 percent of the settlement. Including litigation, Timon spent 57 hours on the case. Chin Lee Florist was awarded a settlement of \$27,000. What did Chin Lee Florist receive (net) as its settlement?
- a. \$27,000
 - b. \$22,950
 - c. \$19,387
 - d. \$15,825

37. Maharan Tent Manufacturing Company's wholesale price for a cabin tent is \$445.00. The profit from the sale of the tent to a retailer is \$15.00. Taxes are \$18.70. Profit and taxes are what percent of the total wholesale price?
- 1.5 percent
 - 5.2 percent
 - 4.8 percent
 - 7.6 percent
38. Drakes Distribution bought two new delivery trucks at a cost of \$40,000 each. The trucks have an estimated life of 5 years and a salvage value of \$15,000 for each truck. What is the book value of both trucks at the end of the first year?
- \$60,500
 - \$72,000
 - \$66,000
 - \$70,000
39. Court Foods estimates the annual cost of carrying inventory to be 25 percent. The company usually has an inventory of \$380,000. What is Court's cost of carrying inventory?
- \$80,000
 - \$95,000
 - \$85,000
 - \$100,000
40. The GDP is \$123.7 million. The population is 300,000. Find the per capita GDP.
- \$412.33
 - \$421.33
 - \$42.23
 - \$433.12
41. A plumber budgeted her \$920 weekly paycheck as follows: $\frac{1}{4}$ for rent, $\frac{1}{3}$ for food, $\frac{1}{6}$ for other expenses, and the remainder to savings. What amount was budgeted for savings each week (to the nearest dollar)?
- \$153
 - \$184
 - \$230
 - \$307
42. If $1\frac{3}{4}$ linear yards of fence material is required to make small animal cages, how many cages can be made from 14 linear yards of the fence material?
- 6
 - 8
 - 10
 - 12

43. What is 65% of 360?
- a. 126
 - b. 234
 - c. 434
 - d. 534
44. It is desired to mix a 20-ounce batch of film developing solution. The instructions state to mix 3 parts developer to 7 parts water. How many ounces of each should be mixed?
- a. 3 ounces developer, 7 ounces water
 - b. 7 ounces developer, 3 ounces water
 - c. 6 ounces developer, 14 ounces water
 - d. 14 ounces developer, 6 ounces water
45. In a graduating class, the ratio of women to men is 7 to 6. If the total graduates is 260, how many are women?
- a. 120
 - b. 140
 - c. 160
 - d. 180
46. If assets are \$98,000 and total liabilities are \$49,000, what is the value of owner's equity?
- a. \$49,000
 - b. \$98,000
 - c. \$147,000
 - d. \$196,000
47. The sum of the angles of a triangle is 180 degrees. Find angles A, B, and C of the triangle if angle A is 31 degrees more than angle B and angle C is 5 degrees less than angle A.
- a. $A = 72$, $B = 41$, $C = 67$
 - b. $A = 41$, $B = 67$, $C = 72$
 - c. $A = 67$, $B = 72$, $C = 41$
 - d. It is impossible to calculate
48. Given the following end-of-the-month information, calculate the reconciled balance for the checking account: Statement ending balance—\$569.20; Outstanding checks—\$51.80 & \$73.10; Service charge—\$6.00; Checkbook balance—\$450.30.
- a. \$444.30
 - b. \$450.30
 - c. \$569.20
 - d. \$575.20

49. Find the perimeter of a circle with a radius of 16 feet. ($\pi = 3.14$)
- a. 50.24
 - b. 75.36
 - c. 100.48
 - d. 252.4
50. A shirt that sells for \$50 has a \$36 cost price. What is the percent markup based on selling price?
- a. 25%
 - b. 28%
 - c. 39%
 - d. 42%